SNAPSMACK OPERABILITY AUDIT — DING DONG BELL

OPAUDIT 019 — The backup that keeps your followers and the exit that forgets them

Audit ID	2026-09-19-019
Report date	2026-09-19
First observed	2026-09-19, early hours. Sean, reading another project's FAQ ("what happens if the relay shuts down?"): "how is the fediverse side of our backups? is it comprehensive? … i think it might be a blindspot or blog only." It was.
Theme	A SnapSmack site is two things — a photo blog and a fediverse account. The raw backup keeps both. The portable "leave with everything" package keeps only the blog. Nobody had asked the second question, so nobody had noticed the answer.
Surfaces	SMACK UP YOUR BACKUP (full dump, RESTORE, exit package); TAKE YOUR SHIT WITH YOU (`core/tyswy-api.php`); the fediverse engine (`core/fediverse.php` : followers, following, blocks, actor keypair, `Move`); DISASTER RECOVERY kit import.
State	**OPEN — PARTLY BUILT** TYSWY 0.2.1 completed the portable fediverse export block. Restore verification, outbound `Move` , and the watched end-to-end lab run remain open.
Watched live?	**Partly built, not yet watched end to end.** TYSWY 0.2.1 now exports the actor, follower/following/block records and importable CSVs. Restore verification and outbound `Move` still require the fediverse test lab.
Reporter	Sean (asked the question) + Claude (measured the answer, wrote the spec).
Related	SECAUDIT 054 (backup signing built then dropped — the user's archive is theirs; this audit is about what the archive *contains*, not who vouches for it). FED UP 719D ("MOVING FROM", the inbound half of what §5 needs). OPAUDIT 017 (delivery pacing the outbound `Move` must reuse).
Publication	Public. No exploitable detail; the gap is a missing feature, not a hole.

1. SUMMARY

Every SnapSmack site that federates has an ActivityPub actor: a keypair, a list of who follows it, who it follows, who it has blocked. That is the part of the site other people are attached to. We promise two things about a site's data: it is backed up, and the owner can leave with it. The first promise is kept in full — the SQL dump takes every `snap_*` table, which includes all of the fediverse tables and the private key. The second is kept for the blog only. TAKE YOUR SHIT WITH YOU exports posts, images, albums, tags, mosaics — and not one follower. And even if it did, a list of followers is not something Mastodon or Pixelfed will read from a file: the only way followers come with you is the ActivityPub `Move` activity, sent by the *old* server, and we can receive one but cannot send one. So the honest state is: leave SnapSmack and your followers stay behind, silently.

2. WHAT A HUMAN SAW

When	What
2026-09-19, ~01:30	Reading holos.social's FAQ: "What happens if the relay shuts down? Your data is safe … point your DNS to another relay." Sean: is our fediverse side of backups comprehensive, or blog only?
~01:40	Measured: the dump is `SHOW TABLES` filtered to `snap_*` — comprehensive. TYSWY's resource list read out: eleven resources, all blog. `Move` grep: inbound handler at `fediverse.php:3213`, `alsoKnownAs` writer at `:3582`, no sender.
~01:45	Sean: "ideally your exports from SMACKONEOUT and GRAMOFSMACK should work for creating a pixelfed account and a SMACKTALK backup should work for mastodon." Checked what those two actually import from outsiders (§4).
~02:00	Sean: "so we have an opaudit because this is another structural failure to address?" Yes.

3. WHAT THE CODE DOES TODAY

Path	Blog content	Followers / following / blocks	Actor keypair	Restore
SUYB full dump	every table	**yes** (`snap_ap_followers`, `snap_ap_following`, `snap_ap_blocks`, `snap_relay_*`, deliveries, likes, boosts, DMs)	**yes** (`snap_settings.fediverse_private_key`)	Database only by hand: DISASTER RECOVERY → import kit. SUYB's RESTORE button puts back **files only** and does not say so.
TYSWY export / exit package	posts, images, post_images, trigrams, mosaics, assets, categories, tags, image_tags, memberships, albums	**none**	no (correct — §5)	—
Outbound `Move`	**does not exist.** Inbound `Move` is honoured with the anti-hijack `alsoKnownAs` check; "MOVING FROM" (719D) lets a site *receive* a migration. Nothing lets a site *send* one.			

4. WHAT THE OTHER SIDE WILL ACTUALLY ACCEPT

This is the part that turns “export a JSON” into something real. Nobody imports someone else’s JSON.

- **Followers** — only through ActivityPub `Move`. The new account names the old actor in `alsoKnownAs`; the old server sends `Move` to every follower; each follower’s server re-follows the new account automatically. No file, no CSV, no JSON does this.
- **Who you follow, blocks, mutes, domain blocks** — Mastodon’s Import page takes CSVs in fixed shapes (`following.csv`: “Account address,Show boosts,Notify on new posts,Languages”; `blocked_accounts.csv`; `muted_accounts.csv`; `blocked_domains.csv`). Pixelfed takes the following list the same way. We hold all of this data; we write none of the files.
- **Posts → Pixelfed** — Pixelfed’s “Import from Instagram” reads Instagram’s archive layout (media files + `posts_1.json` with caption and timestamp). A SMACKONEOUT or GRAMOFSMACK site maps onto that one-to-one.
- **Posts → Mastodon** — nothing. Mastodon imports no posts, not even its own archive. A SMACKTALK site’s people can move (above); its posts stay published where they are, or in the exit package. That is Mastodon’s limit and the FAQ must say so rather than imply otherwise.

5. ROOT CAUSE

The exit package was built when the site was only a blog. TYSWY’s resource list is the blog’s data model from the day it was written. Federation arrived later and grew its own tables; nobody went back and asked whether “everything” still meant everything. Same class as OPAUDIT 016 and 018: a thing that was complete once and quietly stopped being complete when the ground moved.

The portable and the raw were treated as the same promise. They are not. The raw dump is for resurrecting *this* site (so the private key belongs in it). The portable package is for becoming someone *else* (so the key does not, and the followers can only be carried by `Move`). Both are needed; only one existed in full.

RESTORE that restores half. SUYB’s RESTORE uploads files and finishes green. The database is a separate, manual step on another page. A site restored that way looks whole and is a dead actor.

6. WHAT WE WILL DO

Full detail in `_spec/SPEC-fediverse-in-backup-and-exit-v0_1.md`. In order:

1. **TYSWY fediverse block** (core): `exit/fediverse/` with actor, followers, following, blocks as JSON *and* the Mastodon-format CSVs, plus a README that says plainly what each platform will and won’t read. The private key is asserted absent from the whole package.
2. **SUYB says the truth about RESTORE** (desktop): either RESTORE also imports the database through DISASTER RECOVERY behind one explicit yes, or its completion message says “files restored; database: import the kit on DISASTER RECOVERY”. Backup summary gains a line: “Fediverse: N followers, N following, actor key present.”
3. **Outbound `Move`** (core, FED UP “MOVING TO”): preflight that the new account lists us in `alsoKnownAs`, then `Move` to every follower through the paced delivery queue; `movedTo` on the actor; step-up gated; one spoke first.

4. **Instagram-archive adapter** for SMACKONEOUT / GRAMOFSMACK → Pixelfed, beside the WP/Ghost adapters; verified against a real Pixelfed import before the FAQ claims it.
5. **Post-restore fediverse health check**: actor resolves, served public key matches the stored private key, follower count — PASS/FAIL in SUYB's log.

7. VERIFIED

Step	State	Evidence
1. `exit/fediverse/` in the leave package	Built 2026-09-19 (`a7784b63`), not yet run against a live site	TYSWY 0.2.1; server `action=fediverse`; engine test writes all nine files from a fake site and walks the whole archive for a private key (none). PHP contract test 13/13.
2. SUYB RESTORE tells the truth about the database	Open (Codex)	—
3. Outbound `Move`	Open	—
4. Instagram-archive export for Pixelfed	Open	—
5. Post-restore fediverse health check	Open	—

Live, nothing yet. The audit closes on one watched run in the fediverse test lab: back up → destroy the database → RESTORE (files + DB) → the actor still resolves and a follow lands; TYSWY export → `following.csv` imports into the lab Mastodon; `pixelfed-import.zip` imports into the lab Pixelfed; MOVING TO re-points a lab follower. Each step recorded here with a date.

8. EYEBALL NEXT

- Every other promise the FAQ makes about "all your data" — comments, likes received, the shared library's post cache — asked the same question: is it in the raw dump, is it in the portable package, and which of the two *should* hold it.
- The DISASTER RECOVERY import itself has never been watched restoring a federating site. It is the only database restore path we have.

9. THE DOCTRINE THIS INCIDENT CREATED

"Everything" is re-measured every time the data model grows. When a new table family lands (federation, relay, community), the same commit answers: raw dump — yes by construction; portable package — yes or a written reason why not. A backup that keeps the blog and forgets the people is not a backup of a fediverse site.